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FURTHER MATHEMATICS

9231/33

Paper 3 Further Mechanics

May/June 2021

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

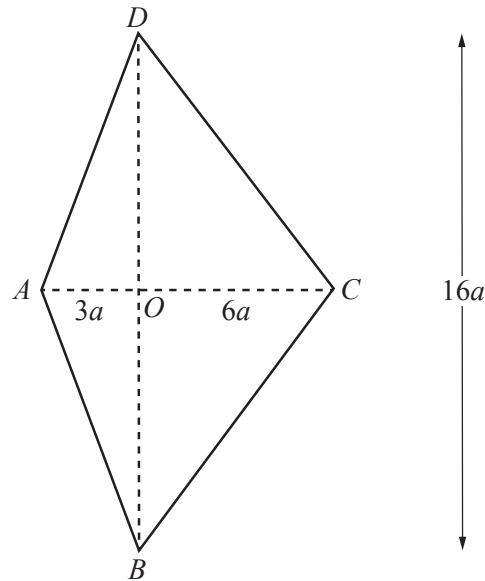
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

1



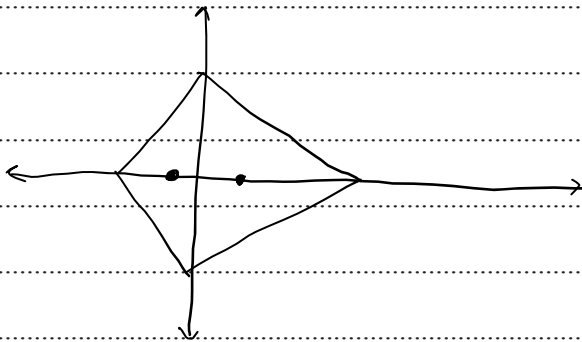
A uniform lamina $ABCD$ consists of two isosceles triangles ABD and BCD . The diagonals of $ABCD$ meet at the point O . The length of AO is $3a$, the length of OC is $6a$ and the length of BD is $16a$ (see diagram).

Find the distance of the centre of mass of the lamina from DB .

[3]

As the shape is symmetric about AC

↳ COM lies on AC



$$(2a)(16a)(\frac{1}{2})(\frac{1}{2})(\rho) + (-a)(16a)(3a)(\frac{1}{2})(\rho) = \bar{x}(16a)(9a)(\frac{1}{2})(\rho)$$

$$12a - 3a = 9\bar{x}$$

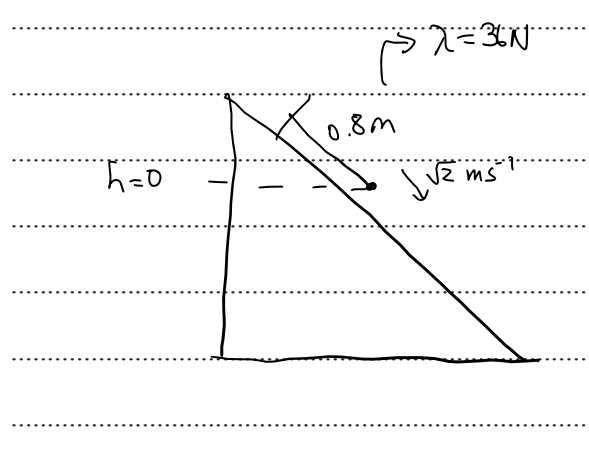
$$9\bar{x} = 9a$$

$$\bar{x} = a \Rightarrow \text{COM is } \boxed{a} \text{ units from } DB$$

- 2 One end of a light elastic string of natural length 0.8 m and modulus of elasticity 36 N is attached to a fixed point O on a smooth plane. The plane is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{3}{5}$. A particle P of mass 2 kg is attached to the other end of the string. The string lies along a line of greatest slope of the plane with the particle below the level of O . The particle is projected with speed $\sqrt{2}\text{ ms}^{-1}$ directly down the plane from the position where OP is equal to the natural length of the string.

Find the maximum extension of the string during the subsequent motion.

[5]



Initially:

$$PE = 0$$

$$KE = \frac{1}{2}(2)(\sqrt{2})^2 = 2\text{ J}$$

$$EPE = 0 \text{ as extension} = 0$$

$\therefore$ Energy of system $= 2\text{ J}$

At maximum extension

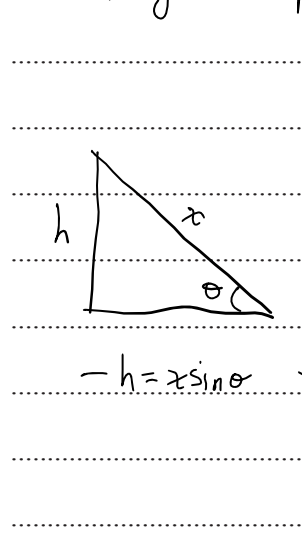
$\hookrightarrow$ particle is turning around

$\hookrightarrow \therefore v = 0$

$\hookrightarrow KE = 0$

$$mgh + \frac{\lambda x^2}{2\lambda} = 2$$

Considering final position



$$\therefore mgh + \frac{\lambda x^2}{2\lambda} = 2$$

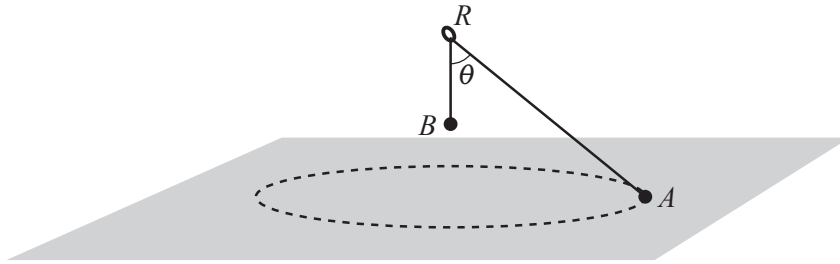
$$-(x)(2)(g)\left(\frac{3}{5}\right) + \frac{36(x)^2}{2(0.8)} = 2$$

$$22.5x^2 - 12x - 2 = 0$$

$$45x^2 - 24x - 4 = 0$$

$$(15x + 2)(3x - 2) = 0$$

$\hookrightarrow$ as $x > 0$, $x = \frac{2}{3}\text{ m}$



Particles A and B , of masses $3m$ and m respectively, are connected by a light inextensible string of length a that passes through a fixed smooth ring R . Particle B hangs in equilibrium vertically below the ring. Particle A moves in horizontal circles on a smooth horizontal surface with speed $\frac{2}{5}\sqrt{ga}$. The angle between AR and BR is θ (see diagram). The normal reaction between A and the surface is $\frac{12}{5}mg$.

(a) Find $\cos \theta$.

[3]

$$T \cos \theta + R = W$$

$$T \cos \theta = 3mg - \frac{12}{5}mg$$

$$T \cos \theta = 0.6mg$$

$$T = B = mg$$

$$\cos \theta = 0.6$$

$$\theta = 53.13^\circ$$

(b) Find, in terms of a , the distance of B below the ring.

[3]

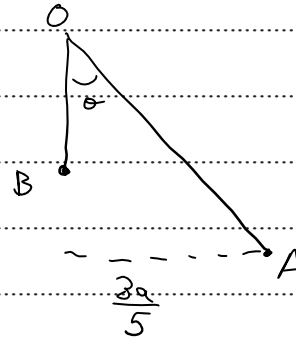
$$T \sin \theta = \frac{mv^2}{r}$$

$$mg \left(\frac{4}{5} \right) = \frac{3m \left(\frac{3\sqrt{5}a}{5} \right)^2}{r}$$

$$\frac{4g}{5} = \frac{12ga}{25r}$$

$$\frac{4r}{5} = \frac{12a}{25}$$

$$r = \frac{3a}{5}$$



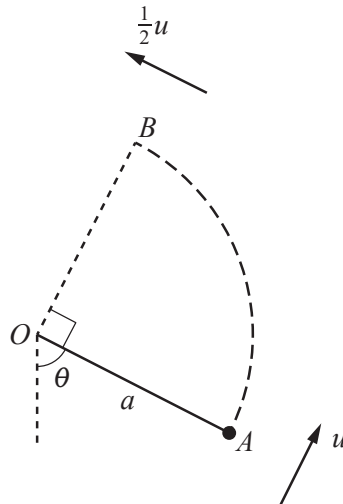
$$\sin \theta = 0.8$$

$$\frac{\frac{3a}{5}}{OA} = 0.8$$

$$OA = \frac{3a}{4}$$

$$OB = a - \frac{3a}{4}$$

$$= \frac{a}{4} \text{ m}$$



A particle of mass m is attached to one end of a light inextensible string of length a . The other end of the string is attached to a fixed point O . The particle is initially held with the string taut at the point A , where OA makes an angle θ with the downward vertical through O . The particle is then projected with speed u perpendicular to OA and begins to move upwards in part of a vertical circle. The string goes slack when the particle is at the point B where angle AOB is a right angle. The speed of the particle when it is at B is $\frac{1}{2}u$ (see diagram).

Find the tension in the string at A , giving your answer in terms of m and g .

[8]

We can derive 4 equations (for tension and force)

↳ Taking initial speed as $n \text{ ms}^{-1}$

$$T_A = \frac{mn^2}{r} - 2mg + 3mg \cos \theta \quad @ A \quad (1)$$

$$T_B = \frac{mn^2}{r} - 2mg + 3mg (\cos(\theta + 90)) \quad @ B \quad (2)$$

$$= \frac{mn^2}{r} - 2mg - 3mg \sin \theta$$

$$\frac{mv^2}{r} = \frac{mn^2}{r} - 2mg + 2mg \cos \theta \quad @ A \quad (3)$$

$$\frac{mv^2}{4r} = \frac{mn^2}{r} - 2mg + 2mg \cos(\theta + 90)$$

$$\frac{mv^2}{r} = \frac{4mn^2}{r} - 8mg - 8mg \sin \theta \quad @ B \quad (4)$$

Equating ③ & ④, we get:

$$\frac{mn^2}{r} - 2mg + 2mg \cos \theta = \frac{4mn^2}{r} - 8mg - 8mg \sin \theta$$

$$0 = \frac{3mn^2}{r} - 6mg - 8mg \sin \theta - 2mg \cos \theta$$

$$0 = \frac{mn^2}{r} - 2mg - \frac{8}{3}mg \sin \theta - \frac{2}{3}mg \cos \theta \quad \textcircled{x}$$

→ we know:

$$T_B = 0 = \frac{mn^2}{r} - 2mg - 3mg \sin \theta \quad \textcircled{y}$$

By comparing ① & ②

$$3mg \sin \theta = \frac{8}{3}mg \sin \theta + \frac{2}{3}mg \cos \theta$$

$$\frac{1}{3} \sin \theta = \frac{2}{3} \cos \theta$$

$$\tan \theta = 2$$

$$\theta = 63.43^\circ$$

$$T_B = 0 = \frac{mn^2}{r} - 2mg - 3mg \sin \theta$$

$$\frac{n^2}{r} = 2g + 3g \sin \theta$$

$$\frac{n^2}{r} = 2g + \frac{6\sqrt{5}}{5}g$$

$$T_A = \frac{mn^2}{r} - 2mg + 3mg \cos \theta$$

$$T_A = m \left(2g + \frac{6\sqrt{5}}{5}g \right) - 2mg + 3mg \left(\frac{1}{5} \right)$$

$$= mg \left(2 + \frac{6\sqrt{5}}{5} - 2 + \frac{3\sqrt{5}}{5} \right)$$

$$= \frac{9\sqrt{5}}{5} mg$$

- 5 A particle P of mass m kg is projected vertically upwards from a point O , with speed 20 ms^{-1} , and moves under gravity. There is a resistive force of magnitude $2mv$ N, where $v \text{ ms}^{-1}$ is the speed of P at time t s after projection.

(a) Find an expression for v in terms of t , while P is moving upwards.

[6]

$$\text{Net force} = -mg - 2mv \quad (\text{upward})$$

$$a = -g - 2v \quad \text{ms}^{-2}$$

$$\frac{dv}{dt} = -g - 2v$$

$$\frac{1}{g+2v} dv = -dt$$

$$\int \frac{1}{g+2v} dv = -\int dt$$

$$\frac{\ln|g+2v|}{2} = -t + c$$

$$\hookrightarrow \text{at } t=0, v=20$$

$$\frac{\ln 50}{2} = c$$

$$\hookrightarrow \frac{\ln|g+2v|}{2} - \frac{\ln 50}{2} = -t$$

Rearranging for v :

$$\ln\left(\frac{g+2v}{50}\right) = -2t$$

$$e^{-2t} = \frac{g+2v}{50}$$

$$g+2v = 50e^{-2t}$$

$$2v = 50e^{-2t} - g$$

$$v = 25e^{-2t} - 5$$

The displacement of P from O is x m at time t s.

- (b) Find an expression for x in terms of t , while P is moving upwards.

[2]

$$v = 25e^{-2t} - 5$$

$$\int v dt = \int 25e^{-2t} - 5 dt$$

$$x = \frac{25e^{-2t}}{-2} - 5t + C$$

$$@ t=0, x=0$$

$$0 = -12.5 + C \Rightarrow C = 12.5$$

$$\hookrightarrow x = -12.5e^{-2t} - 5t + 12.5$$

- (c) Find, correct to 3 significant figures, the greatest height above O reached by P .

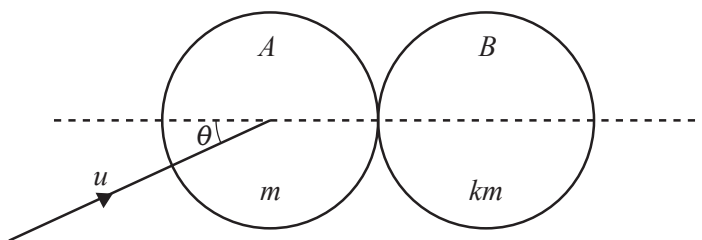
[2]

$$\hookrightarrow \text{greatest height is when } v=0$$

$$\hookrightarrow e^{-2t} = \frac{1}{5}$$

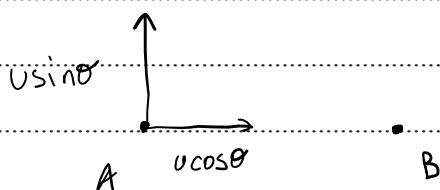
$$x = -12.5 \left(\frac{1}{5} \right) - 5 \left(\frac{\ln 0.2}{-2} \right) + 12.5$$

$$= 5.98 \text{ m}$$



Two uniform smooth spheres A and B of equal radii have masses m and km respectively. Sphere A is moving with speed u on a smooth horizontal surface when it collides with sphere B which is at rest. Immediately before the collision, A 's direction of motion makes an angle θ with the line of centres (see diagram). The coefficient of restitution between the spheres is $\frac{1}{3}$.

- (a) Show that the speed of B after the collision is $\frac{4u \cos \theta}{3(1+k)}$. [3]



CoM: $m u \cos \theta = m v_{xA} + k m v_{xB}$

$u \cos \theta = v_{xA} + k v_{xB}$ (1)

Newton's:

$\frac{v_{xB} - v_{xA}}{u \cos \theta} = \frac{1}{3}$

$v_{xB} - v_{xA} = \frac{u \cos \theta}{3}$ (2)

adding (1) & (2)

$\hookrightarrow (k+1)(v_{xB}) = \frac{u \cos \theta}{3} + u \cos \theta \Rightarrow v_{xB} = \frac{4u \cos \theta}{3(1+k)}$

70% of the total kinetic energy of the spheres is lost as a result of the collision.

(b) Given that $\tan \theta = \frac{1}{3}$, find the value of k .

[6]

$$KE_{\text{initial}} = \frac{1}{2}mv^2$$

$$KE_{\text{final}} = 0.3\left(\frac{1}{2}mv^2\right) = 0.15mv^2$$

$$V_B = \frac{4u \cos \theta}{3(1+k)} = \frac{4u}{\sqrt{10}(1+k)}$$

$$V_{yA} = v \sin \theta = \frac{v}{\sqrt{10}}$$

$$\begin{aligned} V_{xA} &= \frac{4u \cos \theta}{3(1+k)} - \frac{u \cos \theta}{3} = \frac{u \cos \theta}{3} \left(\frac{4}{k+1} - 1 \right) \\ &= \frac{u \cos \theta}{3} \left(\frac{3-k}{k+1} \right) \\ &= \frac{u(3-k)}{\sqrt{10}(k+1)} \end{aligned}$$

$$V_A = \sqrt{V_{xA}^2 + V_{yA}^2}$$

$$= \sqrt{\frac{u^2}{10} + \frac{u^2(k-3)^2}{10(k+1)^2}}$$

$$KE_{\text{final}} = \frac{1}{2}(km) \left(\frac{16u^2}{10(k+1)^2} \right) + \frac{1}{2}m \left(\frac{u^2}{10} + \frac{u^2(k-3)^2}{10(k+1)^2} \right)$$

$$0.30 \cdot \frac{1}{2}mv^2 = \frac{1}{2}km \left(\frac{16u^2}{10(k+1)^2} \right) + \frac{1}{2}m \left(\frac{u^2}{10(k+1)^2} + \frac{u^2(k-3)^2}{10(k+1)^2} \right)$$

$$0.3 = \frac{16k + (k+1)^2 + (k-3)^2}{10(k+1)^2}$$

$$3k^2 + 6k + 3 = 16k + k^2 + 2k + 1 + k^2 - 6k + 9$$

$$k^2 - 6k - 7 = 0$$

$$(k+1)(k-7) = 0$$

$$\text{As } k > 0, \quad \boxed{k=7}$$

- 7 A particle P is projected with speed u at an angle θ above the horizontal from a point O on a horizontal plane and moves freely under gravity. The horizontal and vertical displacements of P from O at a subsequent time t are denoted by x and y respectively.

- (a) Use the equation of the trajectory given in the List of formulae (MF19), together with the condition $y = 0$, to establish an expression for the range R in terms of u , θ and g . [2]

$$x \tan \theta - \frac{g x^2}{2u^2} \sec^2 \theta = 0$$

$$x \left(\tan \theta - \frac{g x}{2u^2} \sec^2 \theta \right) = 0$$

$$\rightarrow \frac{\sin \theta}{\cos \theta} = \frac{g x}{2u^2 \cos^2 \theta}$$

$$x = \frac{2u^2 \sin \theta \cos \theta}{g}$$

$$x = \frac{u^2 \sin 2\theta}{g}$$

- (b) Deduce an expression for the maximum height H , in terms of u , θ and g . [2]

$$\text{if range} = \frac{u^2 \sin 2\theta}{g}$$

$$\rightarrow \text{max H is at } x = \frac{u^2 \sin \theta \cos \theta}{g}$$

$$y = \left(\frac{u^2 \sin \theta \cos \theta \tan \theta}{g} \right) - \frac{g}{2u^2} \left(\frac{1}{\cos^2 \theta} \right) \left(\frac{u^4 \sin^2 \theta \cos^2 \theta}{g^2} \right)$$

$$= \frac{u^2 \sin^2 \theta}{g} - \frac{u^2 \sin^2 \theta}{2g}$$

$$= \frac{u^2 \sin^2 \theta}{2g}$$

It is given that $R = \frac{4H}{\sqrt{3}}$.

(c) Show that $\theta = 60^\circ$.

[1]

$$\frac{2H^2 \sin \theta \cos \theta}{g} = \left(\frac{4}{\sqrt{3}} \right) \left(\frac{H^2 \sin^2 \theta}{2x} \right)$$

$$\cos \theta = \frac{\sin \theta}{\sqrt{3}}$$

$$\tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ$$

It is given also that $u = \sqrt{40} \text{ ms}^{-1}$.

(d) Find, by differentiating the equation of the trajectory or otherwise, the set of values of x for which the direction of motion makes an angle of less than 45° with the horizontal. [4]

$$\frac{dy}{dx} = \tan \theta = \frac{gx}{u^2} \sec^2 \theta$$

$$= \sqrt{3} - \frac{10x}{40} \left(\frac{1}{(\frac{1}{\sqrt{3}})} \right)$$

$$= \sqrt{3} - x$$

$$-1 < \sqrt{3} - x < 1$$

when

$$-1 - \sqrt{3} \leq -x \leq 1 - \sqrt{3}$$

$$\sqrt{3} - 1 \leq x \leq \sqrt{3} + 1$$

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